# MBA CASE STUDY

## Amazon Logistics: Operational Excellence at Unprecedented Scale

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| Company | Amazon Logistics (Amazon.com, Inc.) |
| Industry | E-commerce / Supply Chain / Logistics / Technology |
| Founded | 1994 by Jeff Bezos (logistics transformation began ~2005 with Amazon Prime) |
| Headquarters | Seattle, Washington, USA |
| Key Metrics | 4,000+ delivery stations I 1M+ employees I 200M+ Prime members I 1-day delivery standard I $60B+ annual shipping cost I 500,000+ Kiva robots |
| Case Focus | Operations & Supply Chain Management |
| Teaching Objective | Analyze the operational trade-offs of Amazon's logistics model: speed vs. cost, automation vs. labour, efficiency vs. resilience, profit vs. sustainability. |

### Synopsis

Amazon didn't just build an online bookstore — it built the most sophisticated logistics operation in human history. 500,000 Kiva robots glide through fulfilment centres the size of 28 football fields. 4,000 delivery stations blanket the globe. Predictive shipping algorithms dispatch items before you click "buy." Same-day delivery now reaches 90+ metropolitan areas. The result: 200 million Prime members who expect anything, anywhere, within 48 hours. But this machine runs on relentless efficiency that critics call exploitative — 150% warehouse injury rates above industry average, carbon emissions rivaling a mid-sized country, and competitive practices that regulators on three continents are investigating. This case examines the central trade-offs: speed versus humanity, efficiency versus resilience, profit versus planet.

### Background

From a garage in Bellevue, Washington (1994), Jeff Bezos built Amazon into the world's largest online retailer. The logistics transformation began in earnest with the 2005 launch of Amazon Prime — a $79/year subscription for unlimited free 2-day shipping. Industry analysts called it financial suicide: shipping a $20 book for free, in two days, could not possibly turn a profit. Bezos bet that scale and loyalty would flip the equation, and for years the programme haemorrhaged money before reaching critical mass.

Amazon's warehouse network evolved in three phases. Phase 1 (1997-2005): Single national distribution centres — low complexity, slow delivery to distant regions. Phase 2 (2005-2015): Regional fulfilment centres optimised for inventory placement using predictive demand modelling. Phase 3 (2015-present): Hyper-local sortation centres, same-day facilities, and the launch of AMZL (Amazon Logistics) — a proprietary delivery network that reduced reliance on UPS and FedEx. In 2012 Amazon acquired Kiva Systems for $775M, rebranding it as Amazon Robotics, and today over 500,000 drive units move shelving pods to stationary pickers, cutting "walk time" by 60%.

### The Challenge

Amazon enters 2025+ facing a convergence of pressures. Labor unrest has intensified — the 2022 Staten Island unionisation vote (the first successful US warehouse union) and ongoing organising at facilities in Alabama, Germany, and the UK signal that the workforce is pushing back against 10-hour shifts, injury rates 80% above warehouse industry averages, and algorithmic performance management. Environmental scrutiny is sharpening: Amazon's 2023 carbon footprint exceeded 71 million metric tons, and the company's "Shipment Zero" pledge (net-zero by 2040) faces widespread scepticism given double-digit annual delivery-volume growth. Meanwhile, fuel costs remain volatile, regulators in the EU and US are pursuing antitrust actions, and well-funded competitors (Walmart, Shopify's fulfilment network, Shein/Temu's ultra-cheap air freight model) are eroding Amazon's delivery-speed advantage. Can Amazon sustain its operational dominance while reconciling these external pressures?

### Key Data: Amazon Logistics at a Glance

| Year | Delivery Stations | Prime Members (M) | Shipping Cost ($B) | Robots Deployed |
|---|---|---|---|---|
| 2015 | ~200 | ~50 | $11.5 | 30,000 |
| 2018 | ~500 | ~100 | $27.7 | 100,000 |
| 2020 | ~1,500 | ~150 | $61.1 | 200,000 |
| 2022 | ~2,500 | ~200 | $83.5 | 350,000 |
| 2025 | 4,000+ | 200+ | $60B+ | 500,000+ |

## Discussion Questions

### Supply Chain Design & Optimization

• How did Amazon's decision to build AMZL (Amazon Logistics) rather than relying solely on UPS/FedEx change its cost structure and competitive position?

• What role does predictive analytics play in Amazon's inventory placement strategy? Evaluate the trade-offs between regional vs. national distribution networks.

• How does Amazon's "hub-and-spoke" model compare to traditional logistics networks? Identify one operational bottleneck and propose a solution.

### Automation vs. Human Labor

• What tasks should Amazon automate next, and which tasks should remain human? Use a cost-benefit framework to justify your answer.

• How do Kiva robots change warehouse worker productivity — and injury rates? Research the safety record of automated vs. traditional fulfilment centres.

• If Amazon fully automates its sortation centres, what happens to the 1M+ workforce? Propose a retraining or transition plan.

### Last-Mile Delivery Economics

• The last mile represents up to 53% of total shipping cost. Evaluate Amazon's Flex, DSP, and drone programmes as solutions. Which is most scalable?

• Compare the unit economics of same-day delivery vs. standard 2-day delivery. At what order density does same-day become profitable?

• How would a $2/package carbon surcharge affect the last-mile cost model?

### Sustainability & Resilience

• Amazon's "Shipment Zero" goal targets net-zero carbon by 2040. Is this realistic given growth projections? Critique the pledge using operational data.

• COVID-19 exposed single-source supply chain risks. How should Amazon rebalance "Just-in-Time" efficiency vs. "Just-in-Case" resilience?

• If you were required to cut shipping emissions by 30% in 3 years, what three operational changes would you make first?

## Key Frameworks for Analysis

• Lean Operations: Eliminate waste (muda) in picking, packing, and routing — Amazon's single-piece flow vs. batch processing.

• Six Sigma / DMAIC: Reduce defect rate in order accuracy and delivery windows. Amazon's target: <0.1% error rate.

• Just-in-Time (JIT) vs. Just-in-Case (JIC): The tension between lean inventory and supply chain shocks. Amazon's hybrid approach post-COVID.

• Supply Chain Resilience Framework (Rice & Caniato, 2003): Redundancy, flexibility, and cultural change — mapping Amazon's network on these axes.

• Triple Bottom Line (Elkington, 1994): People, Planet, Profit. Evaluating Amazon's logistics through the lens of social, environmental, and financial performance.

## Teaching Objectives

• Understand the operational design principles behind Amazon's fulfilment network.

• Analyse trade-offs between centralised vs. decentralised distribution models.

• Evaluate the ROI of automation investments (Kiva, drone, autonomous delivery).

• Assess the human and environmental costs of speed-optimised supply chains.

• Apply lean operations and Six Sigma frameworks to a real-world logistics case.

• Develop actionable recommendations for a company facing operational, ethical, and regulatory pressures.

CENTRAL DILEMMA

> Amazon achieves delivery speeds that delight consumers but faces allegations of worker exploitation, environmental damage, and monopoly power. If you were CEO, how would you balance operational efficiency with ethical responsibility — knowing that every concession to workers or the planet could raise prices and slow delivery for 200 million customers?

## Saudi Arabia / Hail Application

• Food delivery logistics for 5-10 restaurant chains in Hail: route clustering, delivery-window optimisation, and driver allocation algorithms.

• Inventory management for bakeries and cafes: applying economic order quantity (EOQ), demand forecasting, and waste-reduction techniques inspired by Amazon's shelf-life modelling.

• Last-mile delivery in a small-city context: how Hail's lower density affects unit economics compared to Riyadh or Jeddah. Opportunity for shared-kitchen ("cloud kitchen") models.

• Waste reduction in food supply chains: cold-chain integrity, FIFO enforcement, and dynamic pricing of near-expiry items — lean principles scaled down for SMEs.

• Vision 2030 logistics hub ambitions: Hail's strategic location on the north-south corridor. Potential to become a regional distribution node for northern Saudi Arabia.

## Pre-Class Assignment

1. Read the case carefully and summarise Amazon's logistics operating model in one paragraph.
2. Choose one discussion-question area (Supply Chain Design, Automation, Last-Mile, or Sustainability) and prepare a 3-minute analysis with supporting data.
3. Research a local Hail food business (restaurant, bakery, or grocer). Map its supply chain from supplier to customer, identify one operational bottleneck, and propose a lean-inspired improvement.
4. Come prepared to debate the central dilemma: should Amazon prioritise speed or ethics? Be ready to defend either side.